

Code

```
#include <SoftwareSerial.h>

SoftwareSerial BT(10, 11); // RX, TX

#define LIGHT1 2
#define LIGHT2 3
#define LIGHT3 4
#define LIGHT4 5
#define FAN 6

char data;

void setup()
{
  pinMode(LIGHT1, OUTPUT);
  pinMode(LIGHT2, OUTPUT);
  pinMode(LIGHT3, OUTPUT);
  pinMode(LIGHT4, OUTPUT);
  pinMode(FAN, OUTPUT);

  // OFF initially
  digitalWrite(LIGHT1, HIGH);
  digitalWrite(LIGHT2, HIGH);
  digitalWrite(LIGHT3, HIGH);
  digitalWrite(LIGHT4, HIGH);
  digitalWrite(FAN, HIGH);

  BT.begin(9600);
}

void loop()
{
  if (BT.available())
  {
    data = BT.read();

    switch(data)
    {
      case 'A':
        digitalWrite(LIGHT1, LOW);
        break;
```

```
case 'a':  
    digitalWrite(LIGHT1, HIGH);  
    break;
```

```
case 'B':  
    digitalWrite(LIGHT2, LOW);  
    break;
```

```
case 'b':  
    digitalWrite(LIGHT2, HIGH);  
    break;
```

```
case 'C':  
    digitalWrite(LIGHT3, LOW);  
    break;
```

```
case 'c':  
    digitalWrite(LIGHT3, HIGH);  
    break;
```

```
case 'D':  
    digitalWrite(LIGHT4, LOW);  
    break;
```

```
case 'd':  
    digitalWrite(LIGHT4, HIGH);  
    break;
```

```
case 'E':  
    digitalWrite(FAN, LOW);  
    break;
```

```
case 'e':  
    digitalWrite(FAN, HIGH);  
    break;
```

```
case 'X': // All ON  
    digitalWrite(LIGHT1, LOW);  
    digitalWrite(LIGHT2, LOW);  
    digitalWrite(LIGHT3, LOW);  
    digitalWrite(LIGHT4, LOW);  
    digitalWrite(FAN, LOW);  
    break;
```

```
case 'x': // All OFF
    digitalWrite(LIGHT1, HIGH);
    digitalWrite(LIGHT2, HIGH);
    digitalWrite(LIGHT3, HIGH);
    digitalWrite(LIGHT4, HIGH);
    digitalWrite(FAN, HIGH);
    break;
}
}
}
```